PYTHON BASIC PROGRAMS

Program 1: Variables and Data Types

```
name = input("Enter your name: ")
age = int(input("Enter your age: "))
percentage = float(input("Enter your percentage: "))
is_student = True
print("\nStudent Details")
print("Name:", name)
print("Age:", age)
print("Percentage:", percentage)
print("Is student:", is_student)
print("\nData types:")
print("name:", type(name).__name__)
print("age:", type(age).__name__)
print("percentage:", type(percentage).__name__)
print("is_student:", type(is_student).__name__)
```

Program 2: Check Whether a Number Is Even or Odd

```
try:
    number = int(input("Enter an integer: "))
    if number % 2 == 0:
        print(number, "is even.")
    else:
        print(number, "is odd.")
except ValueError:
    print("Please enter a valid integer.")
```

Program 3: Factorial of a Number

```
def factorial(number):
    if number < 0:
```

```
        return None
    result = 1
    for value in range(1, number + 1):
        result *= value
    return result
try:
    number = int(input("Enter a non-negative integer: "))
    answer = factorial(number)
    if answer is None:
        print("Factorial is not defined for negative integers.")
    else:
        print(f"{number}! = {answer}")
except ValueError:
    print("Please enter a valid integer.")
```

Program 4: Check Whether a Number Is Prime

```
def is_prime(number):
    if number < 2:
        return False
    for divisor in range(2, int(number ** 0.5) + 1):
        if number % divisor == 0:
            return False
    return True
try:
    number = int(input("Enter an integer: "))
    if is_prime(number):
        print(number, "is a prime number.")
    else:
        print(number, "is not a prime number.")
```

```
except ValueError:
  print("Please enter a valid integer.")
```

Program 5: Multiplication Table

```
try:
  number = int(input("Enter a number: "))
  limit = int(input("How many multiples should be displayed? "))
  if limit < 1:
    print("The limit must be at least 1.")
  else:
    for i in range(1, limit + 1):
      print(f"{number} x {i} = {number * i}")
except ValueError:
  print("Please enter valid integers.")
```

Program 6: Fibonacci Sequence

```
try:
  terms = int(input("Fibonacci terms : "))
  if terms < 1:
    print("Enter a positive number of terms.")
  else:
    first, second = 0, 1
    sequence = []
    for _ in range(terms):
      sequence.append(first)
      first, second = second, first + second
    print("Fibonacci sequence:", sequence)
except ValueError:
  print("Please enter a valid integer.")
```

Program 7: Student Grade Calculator

```
try:
    marks = float(input("Enter marks (0-100): "))
    if not 0 <= marks <= 100:
        print("Marks must be between 0 and 100.")
    elif marks >= 90:
        print("Grade: A")
    elif marks >= 80:
        print("Grade: B")
    elif marks >= 70:
        print("Grade: C")
    elif marks >= 60:
        print("Grade: D")
    else:
        print("Grade: F")
except ValueError:
    print("Please enter a valid number.")
```

DATA STRUCTURES PROGRAMS

Program 1: List Operations

```
numbers = [10, 20, 30, 40, 50]
print("Original list:", numbers)
numbers.append(60)
print("After append(60):", numbers)
numbers.insert(1, 15)
print("After insert(1, 15):", numbers)
numbers.remove(30)
print("After remove(30):", numbers)
```

```
print("First element:", numbers[0])
print("Last element:", numbers[-1])
print("Length:", len(numbers))
print("Sum:", sum(numbers))
print("Minimum:", min(numbers))
print("Maximum:", max(numbers))
numbers.sort()
print("Sorted list:", numbers)
```

Program 2: Remove Duplicates While Preserving Order

```
numbers = [10, 20, 10, 30, 20, 40, 50, 30]
unique_numbers = list(dict.fromkeys(numbers))
print("Original list:", numbers)
print("List without duplicates:", unique_numbers)
```

Program 3: Count Word Frequency

```
from collections import Counter
import re
sentence = input("Enter a sentence: ").lower()
words = re.findall(r"\b\w+\b", sentence)
frequency = Counter(words)
print("\nWord frequencies:")
for word, count in frequency.most_common():
    print(f"{word}: {count}")
```

Program 4: Set Operations

```
python_students = {"Anu", "Bala", "Charan", "Divya"}
ai_students = {"Charan", "Divya", "Esha", "Farhan"}
print("Python students:", python_students)
print("AI students:", ai_students)
```

```
print("Students in both courses:", python_students & ai_students)
print("Students in either course:", python_students | ai_students)
print("Python-only students:", python_students - ai_students)
print("Different students in either set:", python_students ^ ai_students)
```